

# Car Price Prediction using Machine Learning

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## Abstract

This project develops a machine learning model to estimate the selling price of used cars using historical vehicle attributes. The workflow includes data preprocessing, exploratory data analysis, feature engineering, model training, evaluation, and model persistence. Four regression algorithms were compared, with Gradient Boosting Regressor providing the best predictive performance.

## Introduction

Pricing used vehicles accurately is important for buyers, sellers, and dealerships. Machine learning can discover relationships among vehicle characteristics and historical selling prices to support consistent valuation.

## Problem Statement

Predict the selling price of a used car using attributes such as present price, manufacturing year, driven kilometers, fuel type, seller type, transmission, and owner history.

## Objectives

- Understand the dataset.
- Perform EDA.
- Build regression models.
- Compare model performance.
- Save the best model for future predictions.

## Dataset

The dataset contains 301 used-car records with features including Car\_Name, Year, Present\_Price, Driven\_kms, Fuel\_Type, Seller\_Type, Transmission, Owner, and the target variable Selling\_Price.

## Methodology

The workflow consisted of loading the dataset, cleaning data, engineering a Car\_Age feature, encoding categorical variables using one-hot encoding, splitting data into training and testing sets, training multiple regression models, evaluating each model, tuning the Random Forest model with RandomizedSearchCV, and saving the best-performing model.

## Data Preprocessing

The Year column was converted into Car\_Age using the current project year (2026). Car\_Name was removed because it did not contribute meaningful predictive information. Categorical variables were transformed using one-hot encoding.

## Exploratory Data Analysis

EDA included distribution plots, correlation heatmap, box plot by fuel type, feature importance, actual vs predicted comparison, and residual analysis. These visualizations helped understand feature relationships and model behaviour.

## Models Used

1. Linear Regression
2. Decision Tree Regressor
3. Random Forest Regressor
4. Gradient Boosting Regressor

## Model Evaluation

Performance was measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and  $R^2$  Score.

## Performance Comparison

| Model             | MAE    | RMSE   | $R^2$  |
|-------------------|--------|--------|--------|
| Linear Regression | 1.2164 | 1.8658 | 0.8487 |
| Decision Tree     | 0.6966 | 1.0454 | 0.9526 |
| Random Forest     | 0.6369 | 0.9664 | 0.9595 |
| Gradient Boosting | 0.5899 | 0.9394 | 0.9617 |

## Best Model

Gradient Boosting Regressor achieved the best performance with an  $R^2$  score of 0.9617.

## Conclusion

The developed model provides accurate selling price predictions for used cars and demonstrates a complete machine learning workflow.